PART II: PATTERNS

Topic 9: Cycles vs. Seasonality

The Core Concept

- **Seasonality:** Fixed and predictable. Repeats every 12 months (or every week, quarter, etc.).
- **Cycles:** Not fixed. Can last 3–10 years. Driven by broad forces such as economic conditions, social movements, or multi-year reform efforts.

Why It Matters

Confusing a cycle with seasonality leads to bad predictions. If a city enters a 5-year economic downturn, crime may rise every year regardless of the season. If you treat this cycle like a seasonal pattern, you will be confused when numbers do not drop in winter as expected.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

Seasonality in this data: The summer peak (June–July) that repeats every year in Precinct North.

Cycle in this data: The overall decline from 2021 to 2023 — likely reflecting a multi-year reform effort or a broader societal shift — is a cycle. It operates on a scale of years, not months.

These two patterns overlap. The data can fall in summer (seasonal) while also declining year-over-year (cyclical). Analysts must mentally "layer" these patterns.